

LLM Workflows For Applied Research

Empirical Methods — Week 14

Teaching deck draft

Week 14 question

- How can language models support applied research without weakening reproducibility?
- Lecture 14 is about LLMs as research infrastructure, not another text classifier.
- The central object is the workflow: prompt, model, retrieval, schema, benchmark, review, and audit trail.
- The standard remains applied economics: define the object, validate the measure, preserve the replication path.

Five roles for LLMs

Role	Output	Main evidentiary question
Research assistant	code, summaries, templates	Did the researcher verify the final artifact?
Coder / extractor	labels, fields, scores	Does it measure the intended economic object?
Synthetic annotator	scalable labels	Does it agree with a benchmark and where does it fail?
Synthetic subject	choices, survey answers	Does it reproduce human patterns on benchmark tasks?
Simulation engine	actions, transitions, narratives	Does the environment generate disciplined comparative statics?

The role determines the validation standard.

LLMs as research assistants

- High-value uses: code drafting, debugging, cleaning plans, document triage, extraction templates, literature maps.
- Strongest when the final published object is still researcher-controlled.
- Agentic workflows add tool calls, file edits, and intermediate artifacts.
- Every multi-step assistant workflow needs checkpoints and logs.

Assistant use is not automatically a measurement problem, but it is still a reproducibility problem.

LLMs as coders and extractors

$$\widehat{z}_i = g_{\theta}(x_i, p_i, r_i)$$

$$Y_i = \alpha + \beta \widehat{z}_i + u_i$$

- x_i : raw document, posting, contract, response, or case note.
- p_i : prompt and schema.
- r_i : retrieval context, examples, rubrics, or taxonomies.
- θ : model version, decoding, tools, and post-processing.

If $\widehat{z}_i$ enters the economics, the workflow is a measurement instrument.

Synthetic coders and annotators

- Useful when human coding is costly and the construct is well specified.
- Not enough to show that labels look plausible.
- Need human benchmark labels, disagreement audits, and subgroup diagnostics.
- Route ambiguous, high-stakes, and low-confidence cases to review.

A good synthetic-label pipeline is usually human-in-the-loop, not human-free.

LLMs as experimental subjects

$$a_i = s_{\theta}(e_i, p_i, c_i)$$

- e_i : vignette, game, policy scenario, or experimental environment.
- p_i : persona, instructions, ordering, and prompt frame.
- c_i : supplied context or memory.
- a_i : simulated choice, response, message, or action.

Synthetic subjects can pilot designs and explore mechanisms. They do not estimate human behavior unless benchmarked against human behavior.

LLMs as simulation engines

$$x_{t+1} = f_{\theta}(x_t, a_t, \varepsilon_t)$$

Design choice	What to make explicit
State	variables, memory, constraints, histories
Action	feasible choices and timing
Transition	rule-based versus model-generated updates
Shock process	exogenous variation and random seeds
Benchmark	empirical regularities the simulation must reproduce

Fluent narratives are not validation.

Workflow risk decomposition

$$R_{\text{workflow}} = R_{\text{prompt}} + R_{\text{model}} + R_{\text{retrieval}} + R_{\text{benchmark}} + R_{\text{transport}} + R_{\text{privacy}}$$

- Prompt risk: wording changes labels or fields.
- Model risk: versions, providers, or decoding alter outputs.
- Retrieval risk: wrong chunks, stale corpus, missing context.
- Benchmark risk: small, easy, biased, or nonportable gold labels.
- Transport risk: source prompt does not travel to a new domain.
- Privacy risk: raw inputs cannot safely leave the data environment.

Replication, audit trails, and privacy

Artifact	Minimum record
Prompt	system prompt, user template, examples, schema
Model	provider, model, version/date, decoding settings
Retrieval	corpus version, chunking, retrieved passages, settings
Output	raw completion, parsed fields, validation errors
Review	reviewer, adjudication, reason code, final label
Evaluation	benchmark set, metrics, sensitivity scripts
Privacy	release limits, redaction, secure review protocol

Each published value should be traceable from raw input to final dataset.

Publication-ready LLM measurement

- Define the economic object before choosing the model.
- Freeze a schema, prompt, retrieval corpus, and post-processing rule.
- Build or preserve an independent benchmark set.
- Report accuracy, calibration, subgroup performance, and prompt sensitivity.
- Audit false positives, false negatives, unsupported completions, and influential cases.
- Explain how remaining errors change descriptive, causal, structural, or policy interpretation.

Reproduce

- synthetic labor documents
- prompt template
- structured schema
- prompt log
- benchmark metrics

Diagnose

- prompt sensitivity
- subgroup stability
- unsupported outputs
- human-review queue
- downstream slope

Transfer

- workforce case notes
- source prompt
- adapted prompt
- transport metrics
- redesign memo

Takeaways

- LLMs are useful research infrastructure when their role is explicit.
- A prompt/model workflow can be a measurement instrument.
- Synthetic subjects are best treated as pilots or benchmarked objects of study.
- Simulation engines need transparent state, action, transition, and validation rules.
- Replication requires logs, outputs, schemas, benchmarks, review records, and privacy documentation.
- LLMs lower some costs; they do not lower the standard of evidence.